Bro, think of our project as a smart control dashboard for a public transportation operator. The user can monitor buses, manage passenger demand, calculate efficient routes, and simulate bus allocation.

1. User-point-of-view workflow

Step 1: Open the dashboard

The user sees the transport network overview, including the number of bus stops, buses, waiting passengers, and trips.

Step 2: View passenger demand

The system maintains passenger batches with pickup stops, destinations, passenger counts, and arrival order. Larger batches receive higher priority, while FIFO is maintained within the same corridor.

Step 3: Run the simulation

The user clicks Run Simulation. The system selects eligible buses, checks their seating capacity and road restrictions, calculates routes, and assigns passenger batches.

Step 4: Inspect routes and trips

The user can inspect the calculated route and estimated travel time, view assigned trips, and mark individual trips as completed when simulating their arrival.

Step 5: Continue or reset

After trips are completed, buses become available for new allocations. The user can run the simulation again or reset everything to the initial sample scenario.

2. Main features included

- Smart passenger prioritization: Larger passenger batches are considered first, with arrival order breaking ties across eligible batches.
- Capacity-aware bus allocation: Prevents buses from carrying more passengers than their capacity; unserved passengers remain pending.

- Dijkstra route optimization: Finds routes with minimum total travel time.
- Vehicle-specific routing: MiniBuses can use narrow roads that Standard Buses cannot.
- Dynamic traffic updates: Changing a road's travel time can change the shortest route.
- Fleet management: Buses can be marked for maintenance or taken out of service.
- Trip lifecycle management: Trips can be completed, buses can be released, and trip history can be retained.
- Live dashboard and simulation controls: Displays backend data and lets users run, reset, and inspect the simulation.

3. A simple example

Suppose 520 passengers are waiting across 22 batches. When the user runs the simulation, the system might allocate 102 passengers across four buses and leave 418 passengers waiting.

The dashboard then displays the new trip information and remaining demand. After a bus completes its trip, it can be considered for another allocation.

In one sentence: Our application simulates how a transport operator can prioritize passenger demand, allocate buses within capacity limits, and optimize travel routes while tracking the entire process through a dashboard.